MILITARY SPECIFICATION

TANK, MASTER, X-RAY FILM PROCESSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an X-Ray film processing tank complete with inlet and drain hoses and two removable 5 gallon insert tanks suitable for use in Transportable field medical units (MUST).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-7298	Manual, Technical, Commercial Equipment.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6525

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DRAWINGS

Defense Personnel Support Center (DPSC)

24363 Tank, Master, X-Ray Film Processing, Manually Operated, 3 compartment, Capacity 20 gal.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC.
Welders Handbook, 6th Edition

(Application for copies should be addressed to the American Welding Society, Inc., 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Material and construction. Material and construction shall be in accordance with DPSC Drawing 24363 and Bill of Material thereto. The tank shall be supplied with a water inlet hose, drain hose and two 5 gallon insert tanks. A lightweight cover assembly shall also be included and shall be designed as to render the tank light-tight to prevent fogging of photosensitive materials. The tank shall have approximate dimensions of 14-3/4 inches wide by 17-1/2 inches long by 22-3/4 inches deep having a capacity of 20 gallons. Insert tanks of the same size shall be interchangeable among each other. Tank sides shall be equipped with recessed carrying handles. Tanks shall be constructed of corrosion-resisting steel, conforming to class 316 of FED-STD-66 and shall meet the requirements for chemical composition specified therein when tested as specified in 4.4.5.

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3.2 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable filler or reinforcement unless otherwise specified. Fusion through the full thickness of the base metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Fusion through the full thickness of the base metal, in areas other than those where 100 percent penetration is a requirement of joint design, is not acceptable when welding materials over 0.050 inch in thickness. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

3.2.1 Spot welding. All spot welding shall be done with appropriate equipment, properly regulated and operated by thoroughly trained and experienced welders. Spot welds shall be of good quality, free of cracks, porosity, top pickup, blown spots, and other defects which indicate that spot welds were made with dirty electrodes or improperly prepared surfaces. Individual spot welds shall be sufficiently fused to pull the spot out of the joint leaving a hole in one of the welded sheets. Production spot welding shall be checked at the beginning and throughout each run by pulling apart 3/4 inch wide, single-spot, test coupons of material being welded, to determine compliance with the fusion requirements.

3.3 Finish.

3.3.1 Tanks, support frame and cover. The finish of internal and external surfaces of the tanks and support frame shall be commercial No. 2B or No. 4 finish in accordance with FED-STD-66. Any corner filler piece butt welds shall be dressed flush with adjacent surfaces and the top surface of the peripheral edging shall be polished to a No. 4 finish. The dressed surfaces in the vicinity of the welds shall be polished to a No. 4 finish to blend neatly with the internal finish of the tank. Corner welds shall be continuous and dressed to blend smoothly with adjacent exterior surfaces. The exterior surfaces of the cover shall have a No. 4 finish and the interior surfaces a No. 2B or a No. 4 finish.

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3.3.2 Corrosion protection. After fabrication, corrosion-resisting steel components shall be passivated or pickled by complete immersion into a bath of nitric acid or mixture of nitric or hydrofluoric acids of proper concentration. Contamination on surfaces having a No. 4 finish shall be removed without noticeable etching. On surfaces having a No. 2B finish, slight etching will be permitted as a result of the passivation or pickling process to remove concentration. Following the passivation or pickling treatment, components shall be thoroughly washed with water and dried.

3.4 Identification. The master tank shall be permanently and legibly marked in the designated area with the Contract or purchase order number.

3.5 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government. The service data shall be printed on 8-1/2 by 11 inch paper.

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3.6 Workmanship. Workmanship, finish and associated parts shall be first class throughout. The tanks and component parts shall be well made without any operational or other faults and shall be free from defects which detract from their appearance or which may impair their function or serviceability. All exposed corners and edges shall be broken, rounded or otherwise treated to eliminate sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to precribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

Table I.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 for an Inspection level of S-1. Acceptance level shall be zero.

4.3 Examination. The tank shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in Table II.

Table II.

Categories	Defects
Major	
101	Master tank not complete with all component parts.
102	Surfaces not free of burs, sharp edges, and cracked, torn, and deformed areas.
103	Cover fails to fit properly.
104	Mating parts not properly fitted, aligned and interchangeable.
105	Unit not free of corrosion, not properly passivated.

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Table II. (cont'd)

Categories	Defects*
Major	
106	Unit does not rest evenly on flat surface.
107	Welding is not as specified.
Minor	
201	Identification marking not present, correct or complete.
202	Finish not free from scratched, defaced areas, and finish not as specified.
203	Fasteners and rivets not tight, not properly applied.
204	Unit not clean, free of grease, oil, dirt and other foreign matter.
205	Latch handle on cover does not operate properly.

* Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests.

4.4.1 Tank leakage. All welded areas in the sides and bottom of the tanks shall be coated on the outside with a fluorescent penetrant suspended in oil. The welded areas shall be viewed from the inside of the tanks under ultraviolet light in the darkroom for indications of porosity and breaks in the welding. The tanks shall then be filled with water and allowed to stand for 1 hour.

4.4.2 Plug leakage. With the tank insert filled with water and the rubber plug in place, any leakage from the bottom of the insert tank around the rubber plug will be observed and recorded.

4.4.3 Hose and fitting and leakage. With water passing through the drain hose, the hose and connecting fittings shall be observed for leaks.

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4.4.4 Corrosion-resistance test. A 3 to 5 percent salt solution shall be allowed to stand in the tanks for 48 hours. The tanks shall than be examined for evidence of corrosion.

4.4.5 Chemical composition. Tests for chemical analysis shall be conducted in accordance with FED-STD-151. In lieu of the above, a certificate of quality, showing results of tests performed to indicate that material is as specified in 3.1 may be furnished.

4.4.6 Light cover test. The light cover shall be tested as follows to determine compliance with paragraph 3.1:

(a) In total darkness, position unexposed photosensitive material inside of developing tank and resting in inserts and place light cover flush with the top of the tank in a closed position, for normal processing.

(b) A hundred watt lighted incandescent lamp shall be moved around the edges and orifices of the unit for two minutes.

(c) Process photographic material in total darkness in fresh processing solutions at proper temperatures.

(d) Repeat steps (a) and (c). Level of the first batch of photosensitive material shall not exceed that of the second batch of photosensitive material.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table III.

Examination	Defect
Quantity	Quantity in shipping containers not as specified.
Packaging & packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level C as specified (see 6.2).

5.1.1 Level C. Each unit along with accessories shall be packed in a substantial commercial container of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking. Marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Intended for use with a field Medical Unit, Self-contained, Transportable (MUST).

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable level of packing (see 5.1).

6.3 It is noted that this unit is restricted for issue to MUST Field Units.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6525-00-571-9502	TANK, MASTER, X-RAY FILM PROCESSING, Manually Operated 3 Compartment.

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6525-1119
is

☆U.S. GOVERNMENT PRINTING OFFICE: 1975-603-766/954

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-37214 TANK, MASTER, X-RAY FILM PROCESSING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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